

Volume V · Chapter 24 — Perception for Clinical/Surgical Scenes · Theory

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-V-Chapter-24-context.md` for the AI interaction log.

Prerequisites: Ch 23; Ch 6–7 (vision-capable models).

Learning outcomes — the student can: apply computer vision to surgical/clinical scenes; segment/track instruments & anatomy; estimate depth/pose; integrate perception into a robot loop (in simulation).

Topics (theory) — 8 h:

#	Topic	h
24.1	CV for surgical scenes: occlusion, lighting, deformation	2
24.2	Segmentation & instrument tracking	2
24.3	Depth & pose estimation	2
24.4	Integrating perception into control (perception → action)	2

Datasets/tools: OpenCV, segmentation models; the simulator. **Assessment:** perception pipeline (60%); tracking report (20%); quiz (20%). **Key decisions:** model choice vs. latency; sim-to-real gap; **safety of perception errors**. **References:** plan §7; §13 → *Robotics & embodied AI*; *Medical LLMs* (vision). **Hours:** Theory 8 + Lab 14 = 22.